

LQ2 Solutions

B. FALSE

Step-by-step Solution:

1. A differential equation is linear if the dependent variable y and its derivatives appear only to the first power and are not multiplied together.
2. Inspecting the equation $\frac{dy}{dt} = ay - by^2$ reveals the term $-by^2$.
3. Because y is raised to the second power (y^2), the equation is non-linear. Therefore, the statement is True.

Correct Answer: A. TRUE

Q4. What is a reasonable guess for the particular solution of the following ODE?

$$\frac{dy}{dt} = ay + \sin \omega t$$

- A. $A \sin \omega t + B \cos \omega t$, where A, B are arbitrary constants.
- B. $A \sin \omega t$, where A is an arbitrary constant.
- C. $Be^{\cos \omega t}$, where B is an arbitrary constant.
- D. $Ae^{\sin \omega t}$, where A is an arbitrary constant.

Step-by-step Solution:

1. Identify the non-homogeneous forcing term: $f(t) = \sin(\omega t)$.
2. Apply the Method of Undetermined Coefficients: differentiation of $\sin(\omega t)$ generates both sine and cosine terms of the same frequency.
3. Construct the complete trial particular solution as a linear combination of both terms:

$$y_p(t) = A \sin(\omega t) + B \cos(\omega t)$$

where A and B are arbitrary constants.

Correct Answer: A. $A \sin \omega t + B \cos \omega t$, where A, B are arbitrary constants.

Q5. The following ODE is homogeneous, linear, 1st order type. True or False.

$$\frac{dy}{dt} + 2y^2 + 1000 = 0$$

Step-by-step Solution:

1. **Linearity Test:** The presence of the term $2y^2$ makes the ODE **non-linear**.
2. **Homogeneity Test:** Rearrange the equation into standard non-homogeneous form:

$$\frac{dy}{dt} + 2y^2 = -1000$$

The right-hand side has a non-zero constant forcing term (-1000), making it **non-homogeneous**.

3. Because the ODE is neither linear nor homogeneous, the original statement is False.

Correct Answer: B. FALSE